

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)			any two of following observations for (1) each - both release misty or steamy fumes - only sodium bromide gives orange fumes / orange solution - only sodium bromide gives pungent smell (of sulfur dioxide) bromide is more easily oxidised than chloride / sulfuric acid is a strong enough oxidising agent to oxidise bromide but not strong enough to oxidise chloride	2	1		3		2
	(b)	(i)		$K_a = \frac{[H^+][F^-]}{[HF]}$		1		1		
		(ii)		$K_a \times [HF] = [H^+]^2 \quad (1)$ $[H^+] = \sqrt{0.100 \times 7.20 \times 10^{-4}} = 0.00849 \quad (1)$ $pH = -\log [H^+] = 2.07 \quad (1)$		3		3	3	